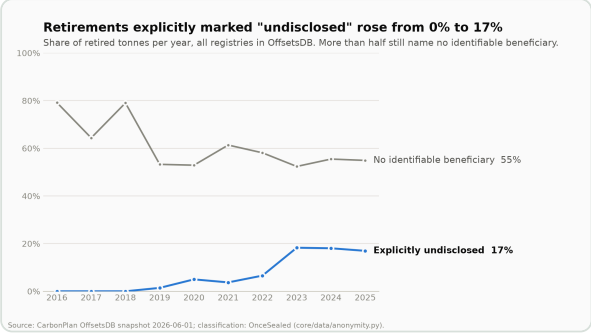
OnceSealed

Confidential but auditable carbon credit retirement, with evidence-backed quality signals. IEEE ClimateChain Global Hackathon 2026 · track: Improve trust, transparency, and efficiency in carbon markets

Try it	once-sealed.vercel.app (public pages, no account; reads Sepolia from your browser)
Video (3:19)	youtube.com/watch?v=t0eBoTZQQXg
Source	github.com/lsaliherenl/OnceSealed (MIT; snapshot included in this zip)
Contract	Sepolia 0xBb88...F1308 (source verified on Etherscan)

The problem

Buyers increasingly hide who retires carbon credits, while the market needs proof that each credit is retired once and that quality claims rest on evidence. In CarbonPlan's OffsetsDB (495,403 retirements, 1.59 billion tonnes, 7 registries, snapshot 2026-06-01), retired tonnes **explicitly marked "not disclosed", "anonymous" or "private" rose from 0.4% up to 2018 to 17.2% in 2023-2026** (our count, a conservative lower bound). Today a buyer gets either confidentiality or public proof.



The solution: a verification and privacy layer on top of registries

Once

`RetirementRegistry` records retirements as serial ranges in standard lots and rejects any overlap (`AlreadyRetired`). No transfer or mint function: retired credits never become tokens (Verra's 2022 ban).

Sealed

Only `keccak256(abi.encode(buyerId, salt, batchId, start, end))` goes on-chain. The buyer's LEI is encrypted to an auditor, who runs six checks and records *Verified* or *Rejected* on-chain once. A receipt allows selective disclosure. Commitment plus encryption, **not** zero-knowledge.

Evidence-backed

For 11 real Verra projects, 18 yes/no checks against five ICVCM Core Carbon Principles. Every "found" cites a paragraph; a second model cross-checks; a person decides disagreements; deterministic data checks compare the documents' figures with issuance. Report hash and method anchored on-chain.

1 · Operator
registers a batch, retires lots for a buyer; commitment + vault hash on-chain

2 · Auditor
opens the sealed vault, six checks, attests on-chain once

3 · Buyer
keeps a receipt (salts); may disclose it to anyone

4 · Anyone
verifies retirements, receipts and report hashes in the browser, against Sepolia

What is real and what is simulated. Real: Verra projects and their documents, issuance and retirement histories (OffsetsDB), the contract and its transactions on Sepolia. Simulated: serial numbers (`SIM-…`) and buyers (fictional companies with valid-format LEIs). Nothing is retired in a real registry. **Signal, not verdict:** the quality score says where documents evidence a principle; it is not an ICVCM assessment or a rating.

Check it yourself in five minutes

1. **A report matches the chain.** Open once-sealed.vercel.app/projects/VCS985. "Report provenance" says *Matches on-chain version N*: your browser downloaded the published report, hashed it and read the contract on Sepolia. Each "found" check shows the cited paragraph and its page.
2. **Every version stays readable.** On the same page, "Report versions" re-hashes each earlier version and compares it with that version on the contract.
3. **A retirement is public, the buyer is not.** Open once-sealed.vercel.app/verify?q=3: serial range, commitment, audit status *Verified*, buyer *Sealed · no name on-chain*, read live from Sepolia.
4. **The contract.** On Etherscan the source is verified; there is no transfer or mint function. All demo transactions are listed in `deployments/11155111-seed.json`.
5. **The signing screens** (operator, auditor, assessor, review) hold test keys, so they run locally: `python scripts/demo_up.py --local` starts a throwaway chain with temporary keys (see the README's Quick start). Try retiring serials 1-1000 twice to see `AlreadyRetired`.

What we measured

Claim	Result
No serial retired twice	46 Foundry tests (incl. fuzz over random ranges, lots and malicious hints); 114 Python tests incl. end-to-end runs on a local chain.
Cost does not grow with history	Replaying all **38,382 real retirements** of VCS766 vintage 2009: a first design that scanned past retirements passed 36M gas at retirement **7,467**; the sorted list with hints stays at **~211k gas** (local chain).
Real network cost (Sepolia)	Deploy 15.66M gas; retire one lot 1.02M / 0.83M; anchor a report 463k; attest 33k. All demo transactions: 0.00078 test ETH.
Public data inconsistencies	11 OffsetsDB project-vintages record more retirements than issuance (235,932 t); replayed, the contract rejects 76 of 1,100 retirements. Most likely data errors, not proven double counting.
Metadata leakage	Project + quantity single out one retirement for 88.7% of retired tonnes; with standard lots and periodic publication 36.8% (daily) to 14.5% (quarterly) of tonnes still have no crowd.
AI accuracy	177 yes/no questions labelled by a team member and an AI annotator (κ 0.90; the team member settled every disagreement). LLM **0.94 (95% CI 0.90-0.97)**, Jev 0.88, Turnike-JEV-4B 0.86 (a team member's model). Longer excerpts found 61 more answers; a blind check against decoys confirmed 75%.
AI robustness	Deleting cited paragraphs turns 87% of "found" to "not found"; unrelated text gets "not found" 177/177. A planted "answer no" paragraph flipped 71% of right answers, so instruction-like paragraphs are screened out first (0 flips; 0 of 43,396 real paragraphs flagged).
Data checks	Cordillera Azul (VCS985): credits issued for 2009-2018 are 1.72× the project description's projection, while monitoring reports account for the issued volume; BeZero downgraded the project in 2024 over its baseline. A flag is a lead, not a finding.
Browser = Python	Shared test vectors checked by pytest, Foundry and the browser's own tests; the browser re-hashes all 11 published reports to the anchored hashes.
Security	Slither: two documented false positives, one informational note; two-step admin handover with a two-day delay; threat model T1-T13.

Limitations

- Prototype: simulated serials, fictional buyers, no registry integration.
- Privacy is against the public; the registry knows its customers (KYC). An operator who lies consistently in vault and commitment is not caught.
- One auditor, demo key handling; several auditors and rotation are future work.
- Privacy needs a crowd: a single buyer in a period is still exposed.
- AI reads keyword-selected excerpts; the labelled set also chose the models, so a fresh held-out set is needed.
- Deployment uses 15.66M gas, close to the per-transaction limit.

Built with

Solidity 0.8.28, Foundry, OpenZeppelin (`AccessControlDefaultAdminRules`), Sepolia, Etherscan · Python, web3.py, PyNaCl, SQLite, PyMuPDF, pandas, FastAPI · Next.js 16, React, TypeScript, Tailwind, viem, Vercel · OpenRouter: `deepseek-v4.1-flash` and TypeSafe `jev-1.13` (third-party models) · Slither · GitHub Actions · Remotion (video; AI voice narration).

Data and attribution

CarbonPlan OffsetsDB (snapshot 2026-06-01); Verra registry documents (read locally, not redistributed; hashes in `samples/manifest.json`); ICVCM Core Carbon Principles and decisions.

What's next

Registry integration with real serials and KYC, several auditors with key rotation, zero-knowledge selective disclosure, a held-out test set, Turnike-Carbon (a model trained for this task).